

Understanding soil and site

Plants have specific requirements and will thrive in the conditions they enjoy most, so taking time to assess your soil type and the amount of sunlight your garden receives will greatly improve your chances of success.

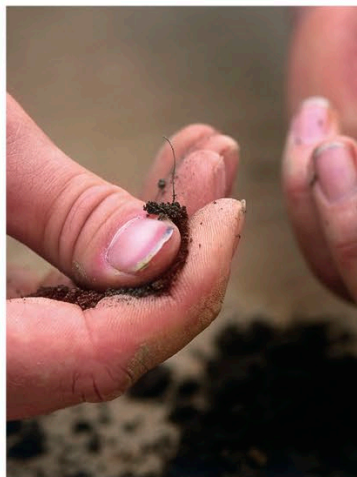
UNDERSTANDING SOILS

Assessing the soil in your garden is the first important step to help you choose the most suitable plants for your conditions. The structure and composition of soil determines how much water it can retain. This also affects its fertility because plants absorb nutrients held in a water solution. Most soils contain a proportion of sand and clay particles, but one type usually dominates. Sand-rich soils tend to be free-draining and have low fertility, due to soluble plants nutrients being washed away.

However, they are also light and easy to dig. Clay soils are heavier and more difficult to cultivate, and although they hold water and nutrients well, they are also prone to waterlogging and cracking when dry. The ideal soil type is loam, which is comprised of roughly equal parts of sand and clay, and is both free-draining and water-retentive. To test your soil type, take samples from around your yard or garden and roll them between your fingers (*see below*).



Acid or alkaline Soils range from acid to alkaline pH, which affects the plants you can grow. Most plants tolerate a range of pH levels but some are particular. Test your soil with an easy-to-use pH testing kit.



Sandy soil When rubbed between your fingers, sandy soil feels gritty, and can't easily be rolled into a ball.



Clay soil Soils rich in clay feel smooth and sticky, and can easily be molded into a ball or ribbon.

IMPROVING SOILS

Many soils benefit from the regular addition of organic matter, such as shredded leaves or garden compost. This improves soil structure, helping sandy soils retain water and nutrients, and improves drainage in clay soils. Dig organic matter into the top 12in (30cm) of soil, or use it as a mulch by applying a thick layer over the surface. For very heavy clay soils, grit can also be added to increase drainage.



Adding organic matter Turn organic matter into the soil with a shovel, pitchfork, or tiller.



Mulching Add a 2in (5cm) layer of well-rotted organic matter over the soil surface each year.



Adding grit Before planting, add small gravel particulates to very heavy clay soils to improve drainage.